StudySphere

Sprint 1 Retrospective (SR1.md)

Date	March 9, 2026
Location	Online (Google Meet)
Sprint	Sprint 1 Retrospective
Project	StudySphere

1. Participants

The Sprint 1 Retrospective meeting was held on March 9, 2026 via Google Meet. All team members attended and participated in the retrospective discussion.

Name	Role	Participated
Benjamin Bradley-Vokey	Developer - Project Setup, Version Control & Branch Integration	Yes
Nimrat Kaur Randhawa	Developer - Join Group	Yes
Navneet Kaur	Developer - Search Groups / System Design Document	Yes
Arshya Baweja	Developer - Authentication & Login	Yes
Gabriel Ottini	Developer - Group Creation	Yes

2. Sprint 1 Summary

Sprint 1 ran from February 23, 2026 to March 9, 2026. The team successfully delivered the core MVP features of StudySphere including user registration, login, study group search, group creation, and join group functionality. All Sprint 1 user stories were completed and demonstrated in the sprint demo video.

User Story	Assigned To	Points	Status
As a student, I want to register my account with username and password, so that I can have persistent data.	Arshya Baweja	3	Completed
As a registered student, I want to login to my account, so that I can access my associated user data.	Arshya Baweja	3	Completed
As a student, I want to search study groups by course code, department and mode, so that I can find relevant academic groups.	Navneet Kaur	5	Completed

As a group leader, I want to be able to create a group for my team to use.	Gabriel Ottini	4	Completed
As a student, I want to join an existing study group, so that I can participate in study sessions.	Nimrat Kaur Randhawa	3	Completed
As a student, I want to be able to change my current courses (Update Profile).	Benjamin Bradley-Vokey	2	Moved to Sprint 2

3. What Went Well

The following practices and outcomes were identified as positive during Sprint 1 and should be continued in Sprint 2:

Good Practices - Sprint 1

- All 5 core Sprint 1 user stories were completed and demonstrated successfully in the demo video
- GitHub branching strategy worked well - each feature was developed on its own branch and merged via pull request
- MongoDB Atlas connected successfully and data was persisted correctly across all features
- Postman was used effectively to test all backend API endpoints before connecting to the React frontend
- Team communication on Discord was consistent throughout the sprint
- System design document was completed with CRC cards, MVC architecture diagram, and error handling table
- bcrypt password hashing was implemented correctly for secure user registration and login
- Search Groups feature supported multiple filters (course code, department, mode) working together
- Join Group correctly prevented duplicate members using $addToSet in MongoDB
- Sprint demo video was recorded and submitted on time before the March 9 deadline

4. What Did Not Go Well

The following issues were identified during Sprint 1 and should be addressed going into Sprint 2:

Areas for Improvement - Sprint 1

- Bug found in createGroup controller - leader was searched using { id: leader } instead of { username: leader } causing leader not found errors
- Frontend was sending currentUser (MongoDB ObjectId) instead of leader username when creating a group - mismatch between frontend and backend

• Duplicate members bug - leader was being added to the members array twice due to both members: [uid] and newGroup.members.push(uid) being called
• Update Profile user story was not completed in Sprint 1 and had to be moved to Sprint 2

• JWT authentication was not implemented in Sprint 1 - login only stores user ID in React state meaning sessions are lost on page refresh
• password field in users.js Mongoose schema had a typo - require: true instead of required: true

• No loading states or proper error messages shown in the frontend UI when API calls fail

• Frontend and backend were not tested together early enough - bugs were only discovered late in the sprint
• Some team members connected to wrong MongoDB database (local vs Atlas), causing data inconsistency between team members

5. Practices to Continue in Sprint 2

The following practices worked well in Sprint 1 and the team agreed to continue them throughout Sprint 2:

Practices to Continue - Sprint 2
• Fix all Sprint 1 bugs before starting new Sprint 2 features - createGroup leader bug, duplicate members, frontend field mismatch
• Implement JWT authentication to keep user sessions active across page refreshes
• Test frontend and backend together earlier in the sprint instead of leaving integration to the end
• Add proper error messages and loading states in the React frontend for all API calls
• Start coding earlier in the sprint to avoid last minute commits before the deadline
• Each team member to post minimum 3 standups on Discord throughout Sprint 2
• Update Trello cards regularly as tasks are completed - not just at the end of sprint
• Complete Update Profile story that was deferred from Sprint 1
• Continue using GitHub branching strategy with feature branches and PR reviews
• Continue testing all backend endpoints in Postman before connecting to frontend

6. New Practices to Adopt in Sprint 2

The following new practices were agreed upon during the retrospective to improve team performance in Sprint 2:

New Practices -Starting Sprint 2
• Post minimum 3 standups per sprint on Discord in the required format - this was not done consistently in Sprint 1
• Test frontend and backend integration together at least 3 days before the sprint deadline
• Use Postman collections shared across the team so everyone can test each other's endpoints
• Code reviews on every pull request before merging - at least one teammate must review and approve
• Update Trello cards daily to reflect actual progress - not just at the end of the sprint
• Write commit messages with ticket numbers (e.g. #18) so commits are linked to Trello cards automatically
• Hold a mid-sprint check-in to catch blockers early instead of finding out at the last standup
• Ensure all team members use the same MONGO_URI pointing to the shared Atlas test database to avoid data inconsistency

7. Harmful Practices to Stop

The following practices were identified as harmful during Sprint 1 and the team agreed to stop them in Sprint 2:

Practices to Stop - Going into Sprint 2
• Leaving bug fixes to the last day of the sprint - bugs found late caused rushed fixes and untested code
• Not testing frontend and backend together until the very end - integration issues should be caught early
• Sending wrong data types between frontend and backend without checking API contracts first (e.g. sending ObjectId instead of username)
• Skipping standup posts - every member must post standups consistently, not just when reminded
• Merging directly to main without a pull request review - all code must go through PR review by at least one teammate
• Leaving Trello cards unchecked even when tasks are done - Trello must reflect real progress at all times
• Each team member using a different database connection string - all must point to the same shared Atlas cluster and test database

8. Best and Worst Experiences of Sprint 1

Best Experience

The best experience of Sprint 1 was successfully completing all 5 core features and recording a working demo video before the March 9 deadline. Seeing the full flow work end to end - from registering an account, logging in, searching for groups, creating a group, and joining a group - was a great team achievement especially for our first sprint together.

Worst Experience

The worst experience was discovering bugs late in the sprint that were caused by mismatches between the frontend and backend - specifically the createGroup controller searching by { id: leader } instead of { username: leader }, and the frontend sending the MongoDB ObjectId instead of the username. These bugs were only caught during final integration testing and caused unnecessary stress close to the deadline.

9. Unfinished Tasks — Carried Over to Sprint 2

The following user stories were not completed in Sprint 1 and have been added to the Sprint 2 backlog. The updated PB.md is saved in doc/sprint2/.

User Story	Original Sprint	Reason Not Completed	Moved To
As a student, I want to be able to update my username and password, so that I can keep my profile information current. (2 pts)	Sprint 1	Team deprioritized to focus on core features - Register, Login, Search, Create, Join Group	Sprint 2 - Assigned to Benjamin Bradley-Vokey
As a registered student, I want my login session to remain active when I navigate the app, so that I do not have to log in every time. (4 pts) - JWT Authentication	Sprint 1	JWT was identified as needed but deferred to Sprint 2 due to time constraints	Sprint 2 - Assigned to Arshya Baweja

Both stories have been added to the Product Backlog (PB.md) and prioritized for Sprint 2. An updated copy of PB.md is saved in doc/sprint2/.

10. Sprint 2 Preview

Based on the Sprint 1 retrospective discussion, the team agreed on the following priorities for Sprint 2:

Feature	Assigned To	Notes
JWT Authentication + Polish Sprint 1 UI + Schedule PDF	Arshya Baweja	Fix security gap from Sprint 1
Update User Info (add/remove courses, change year of study) + Profile UI + Burndown Chart + Merge all PRs	Benjamin Bradley-Vokey	Deferred from Sprint 1 + new profile features
Polish Sprint 1 UI + Session Scheduling UI + User Profile Frontend + My Groups Frontend	Nimrat Kaur Randhawa	UI improvements and frontend polish
Group Session Scheduling (backend + frontend) + Sprint Docs + System Design Doc + CRC Cards + Architecture Diagram + Retrospective + README	Navneet Kaur	New Sprint 2 feature + documentation
My Groups Feature (view all groups user joined) + Location field for group creation	Gabriel Ottini	New Sprint 2 feature (Notification system moved to Sprint 3)